

An Exact Boundary and Skin Atlas for the Finite Hatano–Nelson Chain

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Abstract

We solve the finite asymmetric nearest-neighbor chain across its full hopping cone. Positive open-boundary hopping is diagonally similar to a symmetric path matrix, yielding the exact Chebyshev spectrum. We also obtain a canonical biorthogonal sine basis, distinguish its asymmetry-free density from the right-amplitude skin envelope, and give exact conditioning, propagation, and resolvent formulas.

1 Model and the algebraic core

For $N \geq 2$, let the open-boundary matrix $H_N = H_N(t_R, t_L)$ have $(H_N)_{j,j+1} = t_R$ and $(H_N)_{j+1,j} = t_L$, where $t_R, t_L \geq 0$. Evolution means $i\psi = H_N\psi$. When both hoppings are positive, put

$$g = \sqrt{t_R t_L}, \quad q = \sqrt{t_L/t_R}, \quad D = \text{diag}(1, q, \dots, q^{N-1}),$$

and let A_N be path adjacency. Define

$$\theta_m = \frac{m\pi}{N+1}, \quad S_{jm} = \sqrt{\frac{2}{N+1}} \sin(j\theta_m), \quad 1 \leq j, m \leq N.$$

Theorem 1 (finite-chain atlas). *For $t_R, t_L > 0$,*

$$D^{-1}H_N D = gA_N, \quad \det(zI - H_N) = g^N U_N\left(\frac{z}{2g}\right), \quad E_m = 2g \cos \theta_m. \quad (1)$$

The E_m are real and simple. In the canonical sine gauge,

$$R = DS, \quad L^T = S^T D^{-1}, \quad L^T R = I, \quad \kappa_2(R) = \max(q, q^{-1})^{N-1}. \quad (2)$$

Moreover,

$$e^{-itH_N} = DS \text{diag}(e^{-itE_m}) S^T D^{-1}, \quad (zI - H_N)^{-1} = D(zI - gA_N)^{-1} D^{-1}. \quad (3)$$

For $x = z/(2g)$ and $z \notin \sigma(H_N)$,

$$[(zI - H_N)^{-1}]_{ij} = \frac{q^{i-j} U_{\min(i,j)-1}(x) U_{N-\max(i,j)}(x)}{g U_N(x)}. \quad (4)$$

Proof. Since $D_j^{-1} t_R D_{j+1} = t_R q = g$ and $D_{j+1}^{-1} t_L D_j = t_L/q = g$, diagonal conjugation gives the first identity. Laplace expansion produces the continuant

$$P_0 = 1, \quad P_1 = z, \quad P_N = zP_{N-1} - t_R t_L P_{N-2}.$$

The recurrence for U_N gives (1). The discrete sine transform is orthogonal and diagonalizes A_N , which proves simplicity and reality. Conjugating its columns gives R , while $S^T D^{-1}$ is the exact left dual. Right multiplication by orthogonal S preserves singular values, so the extreme diagonal entries of D give (2). This equality concerns the stated sine gauge; arbitrary column rescaling changes an eigenvector-matrix condition number. Functional calculus gives (3). The standard continuant formula for the inverse of $zI - gA_N$, followed by diagonal conjugation, gives (4). Consequently, for real t ,

$$\|e^{-itH_N}\|_2 \leq \kappa_2(D), \quad \|(zI - H_N)^{-1}\|_2 \leq \frac{\kappa_2(D)}{\text{dist}(z, \sigma(H_N))}. \quad (6)$$

These are bounds, not asserted equalities. $\square$

2 What is, and is not, skin localization

The right and left components in the canonical dual normalization are

$$R_{jm} = q^{j-1} S_{jm}, \quad L_{mj}^T = q^{-(j-1)} S_{jm}.$$

Thus ordinary right amplitudes carry an exponential envelope toward the left for $q < 1$ and toward the right for $q > 1$. Pointwise biorthogonal density is a different object:

$$L_{mj}^T R_{jm} = S_{jm}^2,$$

which is independent of q and sums to one. We use “skin” only for the first, amplitude-level statement.

References

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- [2] N. Hatano and D. R. Nelson, “Non-Hermitian delocalization and eigenfunctions,” *Phys. Rev. B* 58 (1998), 8384–8390. doi:10.1103/PhysRevB.58.8384.